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Editor-in-Chief
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Dear Editor-in-Chief,

I am pleased to submit my manuscript, “*Tiled Dirichlet-Process Gaussian Splatting for UAV Mapping with Calibrated Uncertainty*,” for consideration in IEEE Transactions on Geoscience and Remote Sensing. The paper extends the Dirichlet-process Gaussian-splatting model of my earlier methods paper (arXiv:2607.10912) to survey-scale UAV mapping: independent per-tile variational fits are fused by Hungarian matching under an exact conjugate normal-inverse-Wishart marginal-likelihood score, merged in closed form, and stitched by a partition-of-unity gate into a single *normalized*, provenance-preserving scene-level predictive density. On three Hessigheim 3D LiDAR epochs (59–129 million points) and a second 152-million-point photogrammetric site, the stitched density matches the hard seam-deletion rules of current tiled practice on held-out log-density while deleting nothing, and the resulting calibrated uncertainty is then spent on four mapping tasks: error sparsification, mesh-referenced accuracy stratification, change detection with a model-derived level of detection, and re-flight planning, all on CPU-only laptop hardware.

I believe TGRS is the right home for this work because its contribution is operational rather than a methodological display: tiled processing of large UAV surveys is standard practice in this community, the seam problem it addresses is a practice problem, and every evaluation is a mapping-product evaluation (held-out predictive density, empirical interval coverage, geometric accuracy against the benchmark’s reference mesh, null-calibrated change detection on repeat LiDAR epochs, and acquisition planning) rather than a rendering benchmark. The datasets (Hessigheim 3D, GauU-Scene, Mill 19, STPLS3D) and the tasks are those of the remote-sensing literature the paper engages.

I have tried to make the paper’s claims exactly as strong as its evidence and no stronger, and its controls and ablations are reported at parity: seam deletion is empirically near-costless on these scenes and the paper says so; seam-band parity and seam calibration hold at production capacity on the H3D epochs but do not replicate at the second site, and the abstract carries that caveat; the sparsification signal’s point-density confound is decomposed rather than hidden; the change-detection level of detection has a stated ≈ 1 m representation-limit floor; and the re-flight result is dissected against oracle, random, and mass-only controls, including the regime where the mass-only control matches the planner. A merge ablation and a no-merge refit isolate what

the exact conjugate merge contributes. Complete code, fixed-seed reproduction commands, and machine-readable experiment records accompany the submission.

This manuscript is not under consideration elsewhere; the methods paper it builds on is cited and none of its figures or results are reused. I have no conflicts of interest to declare.

Thank you for your consideration. I look forward to the reviewers' comments.

Sincerely,

Aqi Dong